

Lab Assignment 19.2 – Code Translation

M.Rushikesh

2303A51399

Task 1: Python to Java Conversion (File Handling)

Question

Convert a Python file handling program into Java.

Prompt

"Translate a Python file read/write program into Java with proper file handling and exception handling."

Code

Python Code

```
with open("sample.txt", "w") as f:
```

```
    f.write("Hello World")
```

```
with open("sample.txt", "r") as f:
```

```
    print(f.read())
```

Java Code

```
import java.io.*;
```

```
public class FileExample {
```

```
    public static void main(String[] args) {
```

```
        try {
```

```
            FileWriter writer = new FileWriter("sample.txt");
```

```
            writer.write("Hello World");
```

```
            writer.close();
```

```
            BufferedReader reader = new BufferedReader(new FileReader("sample.txt"));
```

```
            String line = reader.readLine();
```

```
        System.out.println(line);
        reader.close();
    } catch (IOException e) {
        e.printStackTrace();
    }
}
}
```

Output

Hello World

Code Explanation

- Python file handling uses with open()
 - Java uses FileWriter and BufferedReader
 - Exception handling required in Java
 - Both produce same output
-

Task 2: C++ to Python Conversion (Sorting)

Question

Convert C++ sorting program into Python.

Prompt

"Translate C++ bubble sort program into Python and verify correctness."

Code

C++ Code

```
#include <iostream>

using namespace std;

int main() {
```

```
int arr[4] = {5, 2, 9, 1};
for(int i=0;i<4;i++){
    for(int j=0;j<4-i-1;j++){
        if(arr[j] > arr[j+1]){
            swap(arr[j], arr[j+1]);
        }
    }
}
for(int i=0;i<4;i++)
    cout<<arr[i]<<" ";
}
```

Python Code

```
arr = [5, 2, 9, 1]

for i in range(len(arr)):
    for j in range(len(arr)-i-1):
        if arr[j] > arr[j+1]:
            arr[j], arr[j+1] = arr[j+1], arr[j]

print(arr)
```

Output

[1, 2, 5, 9]

Code Explanation

- Same bubble sort logic used
 - Python uses list swapping directly
 - C++ uses swap() function
 - Output remains identical
-
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Task 3: SQL to Pandas (GROUP BY)

Question

Convert SQL aggregation query into Pandas.

Prompt

"Translate SQL GROUP BY query into Pandas groupby() operation."

Code

SQL Query

```
SELECT department, SUM(sales)
FROM employees
GROUP BY department;
```

Python Code

```
import pandas as pd

data = {
    "department": ["HR", "IT", "HR", "IT"],
    "sales": [100, 200, 150, 300]
}

df = pd.DataFrame(data)

result = df.groupby("department")["sales"].sum()
print(result)
```

Output

```
HR    250
IT    500
```

Code Explanation

- SQL GROUP BY → Pandas groupby()

- SUM aggregation applied
 - Same grouped result achieved
-
-

Task 4: Multi-Language Snippet Converter

Question

Convert Python API into another language.

Prompt

"Convert Python Flask API into Node.js Express API."

Code

Python (Flask)

```
from flask import Flask  
app = Flask(__name__)
```

```
@app.route('/')  
def home():  
    return "Hello API"
```

```
app.run()
```

Node.js (Express)

```
const express = require('express');  
const app = express();
```

```
app.get('/', (req, res) => {  
    res.send("Hello API");  
});
```

```
app.listen(3000);
```

Output

Hello API

Code Explanation

- Flask routes → Express routes
 - Python functions → JavaScript callbacks
 - Same API functionality
-

Task 5: Python to C++ (Even/Odd)

Question

Convert Python function to check even/odd into C++.

Prompt

"Translate Python even/odd function into C++ with input handling."

Code

Python Code

```
def check_even_odd(n):  
    if n % 2 == 0:  
        return "Even"  
    else:  
        return "Odd"  
  
print(check_even_odd(4))
```

C++ Code

```
#include <iostream>  
  
using namespace std;  
  
int main() {  
    int n;
```

```
cin >> n;

if(n % 2 == 0)
    cout << "Even";
else
    cout << "Odd";

return 0;
}
```

Output

Input: 4

Output: Even

Code Explanation

- Modulus operator checks divisibility
- Python uses function, C++ uses main
- Same logic implemented
- Works for multiple inputs